전산화학 (2012년 1학기) 2차 시험 해답

(총 **8**문제, **100**점)

1. **(총 10점, 각 5점)**

(a)

$$\frac{dq(T)}{dT} = \frac{E}{k_B T^2} e^{-E/(k_B T)}$$
$$\frac{d^2 q(T)}{dT^2} = -\frac{2E}{k_B T^3} e^{-E/(k_B T)} + \left(\frac{E}{k_B T^2}\right)^2 e^{-E/k_B T}$$

(b)

$$\frac{df(v)}{dv} = -\frac{2mv}{k_B T} e^{-mv^2/(k_B T)}$$
$$\frac{d^2 f(v)}{dv^2} = -\frac{2m}{k_B T} e^{-mv^2/(k_B T)} + \left(\frac{2mv}{k_B T}\right)^2 e^{-mv^2/k_B T}$$

2. **(총 10점, 각 5점)**

(a)

$$f_x(x,y) = \left(\frac{\partial f}{\partial x}\right)_y = \frac{3}{3x+5y}$$
$$f_y(x,y) = \left(\frac{\partial f}{\partial y}\right)_x = \frac{5}{3x+5y}$$
$$f_{xx}(x,y) = \left(\frac{\partial^2 f}{\partial x^2}\right)_y = -\frac{9}{(3x+5y)^2}$$
$$f_{yy}(x,y) = \left(\frac{\partial^2 f}{\partial y^2}\right)_y = -\frac{25}{(3x+5y)^2}$$
$$f_{xy}(x,y) = \left(\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)_y\right)_x = -\frac{15}{(3x+5y)^2}$$

(b)

$$f_x(x,y) = \left(\frac{\partial f}{\partial x}\right)_y = \frac{y}{(x+y)^2}$$
$$f_y(x,y) = \left(\frac{\partial f}{\partial y}\right)_x = -\frac{x}{(x+y)^2}$$
$$f_{xx}(x,y) = \left(\frac{\partial^2 f}{\partial x^2}\right)_y = -\frac{2y}{(x+y)^3}$$
$$f_{yy}(x,y) = \left(\frac{\partial^2 f}{\partial y^2}\right)_x = \frac{(x+y)^2 - 2x(x+y)}{(x+y)^4} = \frac{-x+y}{(x+y)^3}$$
$$f_{xy}(x,y) = \left(\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)_y\right)_x = \frac{(x+y)^2 - 2y(x+y)}{(x+y)^4} = \frac{x-y}{(x+y)^3}$$

3. **(총 20점, 각 10점)**

(a) $f(x,y)$의 임계점을 (x^*,y^*)라 하면, 임계점에서

$$f_x(x^*,y^*) = 0 \Rightarrow 4(x^*)^3 - 4y^* = 0 \Rightarrow y^* = (x^*)^3$$
$$f_y(x^*,y^*) = 0 \Rightarrow 4(y^*)^3 - 4x^* = 0 \Rightarrow x^* = (y^*)^3$$

위 두 식을 연립하면,

$$(x^*)^9 - x^* = 0 \Rightarrow x^*(x^*-1)(x^*+1)((x^*)^2+1)((x^*)^4+1) = 0$$
$$\Rightarrow x^* = 0 \quad \text{또는} \quad x^* = 1 \quad \text{또는} \quad x^* = -1$$

그러므로 **임계점은 (0,0), (1,1), (-1,-1)** 이다. Hessian 검사를 이용하면

$$\Delta_1 = f_{xx}(x^*,y^*) \quad \text{그리고} \quad \Delta_2 = f_{xx}(x^*,y^*)f_{yy}(x^*,y^*) - f_{xy}(x^*,y^*)^2.$$

주어진 함수 $f(x,y) = x^4 + y^4 - 4xy + 1$에 대하여

$$f_{xx}(x,y) = 12x^2$$
$$f_{yy}(x,y) = 12y^2$$
$$f_{xy}(x,y) = -4$$

i. $(0,0)$에서

$$f_{xx}(0,0) = 0$$
$$f_{yy}(0,0) = 0$$
$$f_{xy}(0,0) = -4$$

그러므로

$$\Delta_1 = 0 \quad \text{그리고} \quad \Delta_2 = (0)\times(0) - (-4)^2 = -16 < 0$$

$\Delta_2 < 0$ 이므로 $f(x,y)$는 $(0,0)$에서 안장점을 갖는다.

ii. $(1,1)$에서

$$f_{xx}(1,1) = 12$$
$$f_{yy}(1,1) = 12$$
$$f_{xy}(1,1) = -4$$

그러므로

$$\Delta_1 = (12)^2 = 144 > 0 \quad \text{그리고} \quad \Delta_2 = (12)\times(12) - (-4)^2 = 128 > 0$$

$\Delta_1 > 0$ **이고** $\Delta_2 > 0$ **이므로** $f(x,y)$ **는** $(1,1)$ **에서 극소값을 갖는다.**

iii. $(1,1)$ 에서

$$\begin{aligned} f_{xx}(-1,-1) &= 12 \\ f_{yy}(-1,-1) &= 12 \\ f_{xy}(-1,-1) &= -4 \end{aligned}$$

그러므로

$$\Delta_1 = (12)^2 = 144 > 0 \quad \text{그리고} \quad \Delta_2 = (12)\times(12) - (-4)^2 = 128 > 0$$

$\Delta_1 > 0$ **이고** $\Delta_2 > 0$ **이므로** $f(x,y)$ **는** $(-1,-1)$ **에서 극소값을 갖는다.**

(b) $f(x,y)$ 의 임계점을 (x^*, y^*) 라 하면, 임계점에서

$$\begin{aligned} f_x(x^*,y^*) = 0 &\Rightarrow \frac{2x^*}{(x^*)^2+(y^*)^2+2} = 0 \Rightarrow x^* = 0 \\ f_y(x^*,y^*) = 0 &\Rightarrow \frac{2y^*}{(x^*)^2+(y^*)^2+2} = 0 \Rightarrow y^* = 0 \end{aligned}$$

그러므로 임계점은 (0,0) 이다. Hessian 검사를 이용하면

$$\Delta_1 = f_{xx}(x^*,y^*) \quad \text{그리고} \quad \Delta_2 = f_{xx}(x^*,y^*)f_{yy}(x^*,y^*) - f_{xy}(x^*,y^*)^2.$$

주어진 함수 $f(x,y) = \ln(x^2+y^2+2)$ 에 대하여

$$\begin{aligned} f_{xx}(x,y) &= \frac{2(x^2+y^2+2)-(2x)(2x)}{(x^2+y^2+2)^2} = \frac{-2x^2+2y^2+4}{(x^2+y^2+2)^2} \\ f_{yy}(x,y) &= \frac{2(x^2+y^2+2)-(2y)(2y)}{(x^2+y^2+2)^2} = \frac{2x^2-2y^2+4}{(x^2+y^2+2)^2} \\ f_{xy}(x,y) &= \frac{-4xy}{(x^2+y^2+2)^2} \end{aligned}$$

$(0,0)$ 에서

$$\begin{aligned} f_{xx}(0,0) &= 1 \\ f_{yy}(0,0) &= 1 \\ f_{xy}(0,0) &= 0 \end{aligned}$$

그러므로

$$\Delta_1 = 1 \quad \text{그리고} \quad \Delta_2 = (1)\times(1) - (0)^2 = 1 > 0$$

$\Delta_1 > 0$ **이고** $\Delta_2 > 0$ **이므로** $f(x,y)$ **는** $(0,0)$ **에서 극소값을 갖는다.**

4. **(총 15점, 각 5점)**

(a) $df(x,y) = \tan y dx + x\sec^2 dy$

$$M(x,y) = \tan y, \qquad \text{그리고} \qquad N(x,y) = x\sec^2$$
$$M_y(x,y) = \left(\frac{\partial M}{\partial y}\right)_x = \sec^2 y$$
$$N_x(x,y) = \left(\frac{\partial N}{\partial x}\right)_y = \sec^2 y$$
$$M_y(x,y) = N_x(x,y)$$

완전미분, $f(x,y) = x\tan y + C$

(b) $df(x,y) = \ln y dx + (x/y)dy$

$$M(x,y) = \ln y, \qquad \text{그리고} \qquad N(x,y) = \frac{x}{y}$$
$$M_y(x,y) = \left(\frac{\partial M}{\partial y}\right)_x = \frac{1}{y}$$
$$N_x(x,y) = \left(\frac{\partial N}{\partial x}\right)_y = \frac{1}{y}$$
$$M_y(x,y) = N_x(x,y)$$

완전미분, $f(x,y) = x\ln y + C$

(c) $df(x,y) = (2x/y)dx - (x^2/y^2)dy$

$$M(x,y) = \frac{2x}{y} \qquad \text{그리고} \qquad N(x,y) = \frac{x^2}{y^2}$$
$$M_y(x,y) = \left(\frac{\partial M}{\partial y}\right)_x = -\frac{2x}{y^2}$$
$$N_x(x,y) = \left(\frac{\partial N}{\partial x}\right)_y = -\frac{2x}{y^2}$$
$$M_y(x,y) = N_x(x,y)$$

완전미분, $f(x,y) = x^2/y + C$

5. **(총 10점, 각 5점)**

(a)

$$\int_{V_1}^{V_2}\left(\frac{nRT}{V-nb} - \frac{n^2a}{V^2}\right)dV = \left[nRT\ln(V-nb) + \frac{n^2a}{V}\right]_{V_1}^{V_2},$$
$$= nRT\ln(V_2-b) - nRT\ln(V_1-nb) + \frac{n^2a}{V_2} - \frac{n^2a}{V_1},$$
$$= nRT\ln\left(\frac{V_2-nb}{V_1-nb}\right) + \frac{n^2a}{V_2} - \frac{n^2a}{V_1}.$$

(b) $t = J^2 + J$ 라 하면, $dt = (2J+1)dJ$.

$$\int_0^\infty (2J+1)e^{-a(J^2+J)}dJ = \int_0^\infty e^{-at}dt = \frac{1}{a}$$

6. **(총 10점, 각 5점)**

(a) $\nu \to 0$ 일 때 분자, 분모 모두 0 이 되므로, l'Hôpital 의 법칙을 이용하면,

$$\begin{aligned}\lim_{\nu\to 0}\frac{Nh\nu}{e^{h\nu/(k_BT)}-1} &= \lim_{\nu\to 0}\frac{Nh}{(h/(k_BT))e^{h\nu/(k_BT)}}\\ &= \lim_{\nu\to 0}\frac{Nk_BT}{e^{h\nu/(k_BT)}}\\ &= Nk_BT\end{aligned}$$

(b) $T \to \infty$ 일 때, 분모가 ∞ 가 되므로,

$$\lim_{T\to\infty}\frac{Nh\nu}{e^{h\nu/(k_BT)}-1} = 0$$

7. **(총 10 점)**

$$\int_C((1-ye^{-x})dx + e^{-x}dy \quad \text{여기에서} \quad C:(0,1)\to(1,2) \tag{1}$$

$M(x,y) = 1-ye^{-x}$, 그리고 $N(x,y) = e^{-x}$ 라 놓으면,

$$\left(\frac{\partial M}{\partial y}\right)_x = -e^{-x}, \qquad \left(\frac{\partial N}{\partial x}\right)_y = -e^{-x} \Rightarrow \left(\frac{\partial M}{\partial y}\right)_x = \left(\frac{\partial N}{\partial x}\right)_y$$

(2) 식으로부터 $(1-ye^{-x}dx + e^{-x}dy$ 는 완전미분이다.

(a) (**방법 1**) 그러므로 위 선적분은 경로에 의존하지 않고, 적분하기 쉬운 임의의 경로로 적분할 수 있다. 경로를 $(0,1)\to(1,1)$ 그리고 $(1,1)\to(1,2)$ 로 나누면, (1) 식은

$$\begin{aligned}\int_C(1-ye^{-x})dx + e^{-x}dy &= \int_{(0,1)}^{(1,1)}(1-ye^{-x})dx + e^{-x}dy + \int_{(1,1)}^{(1,2)}(1-ye^{-x})dx + e^{-x}dy\\ &= \int_0^1(1-(1)e^{-x})dx + \int_1^2 e^{-1}dy\\ &= \Big[x+e^{-x}\Big]_0^1 + \Big[e^{-1}y\Big]_1^2\\ &= (1+e^{-1}) - 1 + 2e^{-1} - e^{-1} = \mathbf{2e^{-1}}\end{aligned}$$

(b) (**방법 2**) $df(x,y)$ 가 완전미분이으로 $f(x,y)$ 를 구하면,

$$f(x,y) = x + ye^{-x} + C \tag{2}$$

그러므로 (1) 식은

$$\begin{aligned}\int_C((1-ye^{-x})dx + e^{-x}dy &= \Big[x+ye^{-x}\Big]_{(0,1)}^{(1,2)}\\ &= 1 + 2e^{-1} - (1) = \mathbf{2e^{-1}}\end{aligned}$$

8. **(총 15 점, 각 5 점)**

(a) 연쇄법칙을 이용하면,

$$\left(\frac{\partial z}{\partial r}\right)_\theta = \left(\frac{\partial x}{\partial r}\right)_\theta \left(\frac{\partial z}{\partial x}\right)_y + \left(\frac{\partial y}{\partial r}\right)_\theta \left(\frac{\partial z}{\partial y}\right)_x \tag{3}$$

한편, $x = r\cos\theta$ 이고, $y = r\sin\theta$ 이므로

$$\left(\frac{\partial x}{\partial r}\right)_\theta = \cos\theta, \quad \text{그리고} \quad \left(\frac{\partial y}{\partial r}\right)_\theta = \sin\theta \tag{4}$$

(4) 식을 (3) 식에 대입하면,

$$\boxed{\left(\frac{\partial z}{\partial r}\right)_\theta = \cos\theta\left(\frac{\partial z}{\partial x}\right)_y + \sin\theta\left(\frac{\partial z}{\partial y}\right)_x} \tag{5}$$

(b) (a)와 같은 방법으로

$$\left(\frac{\partial z}{\partial \theta}\right)_r = \left(\frac{\partial x}{\partial \theta}\right)_r \left(\frac{\partial z}{\partial x}\right)_y + \left(\frac{\partial y}{\partial \theta}\right)_r \left(\frac{\partial z}{\partial y}\right)_x \tag{6}$$

한편, $x = r\cos\theta$ 이고, $y = r\sin\theta$ 이므로

$$\left(\frac{\partial x}{\partial \theta}\right)_r = -r\sin\theta, \quad \text{그리고} \quad \left(\frac{\partial y}{\partial \theta}\right)_r = r\cos\theta \tag{7}$$

(7) 식을 (6) 식에 대입하면,

$$\boxed{\left(\frac{\partial z}{\partial \theta}\right)_r = -r\sin\theta\left(\frac{\partial z}{\partial x}\right)_y + r\cos\theta\left(\frac{\partial z}{\partial y}\right)_x} \tag{8}$$

(c) (5) 식과 (8) 식으로부터

$$\sin\theta\left(\frac{\partial z}{\partial x}\right)_y - \cos\theta\left(\frac{\partial z}{\partial y}\right)_x = -\frac{1}{r}\left(\frac{\partial z}{\partial \theta}\right)_r \tag{9a}$$

$$\cos\theta\left(\frac{\partial z}{\partial x}\right)_y + \sin\theta\left(\frac{\partial z}{\partial y}\right)_x = \left(\frac{\partial z}{\partial r}\right)_\theta \tag{9b}$$

(9a) 식과 (9b) 식의 양변을 각각 제곱하여 정리하면,

$$\boxed{\left(\frac{\partial z}{\partial x}\right)_y^2 + \left(\frac{\partial z}{\partial y}\right)_x^2 = \left(\frac{\partial z}{\partial r}\right)_\theta^2 + \frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)_r^2} \tag{10}$$